

KELETSO MAKALELA

Software Developer | Backend Developer | AI & Automation Enthusiast

Gauteng Johannesburg, South Africa

073 749 0771

Keletso.makalela512@gmail.com

keletsomakalela@icloud.com

GitHub: <https://github.com/KeletsoNIT>

Portfolio: <https://keletsonit.github.io/Portfolio/>

PROFESSIONAL SUMMARY

Results-driven Software Developer with a Bachelor of Science in Information Technology and hands-on experience developing full-stack applications, backend systems, and AI-powered software solutions. Recently completed an **Agentic AI Software Development Internship**, where I gained practical experience designing intelligent AI systems, building agent-based applications, and applying modern AI development frameworks to solve real-world business problems.

Experienced in designing scalable applications using Python, React, Node.js, Flask, JavaScript, SQL, REST APIs, Docker, and Git while following Agile software development practices. Demonstrated the ability to transform complex business requirements into practical software solutions through the development of an AI-powered Recruitment Agent capable of CV parsing, semantic candidate matching, automated email communication, PDF report generation, and multi-agent recruitment workflows.

Passionate about artificial intelligence, backend development, automation, and clean software architecture, with a strong commitment to continuous learning and delivering high-quality, maintainable software. Seeking an opportunity as a Software Developer,

Backend Developer, Full-Stack Developer, or AI Developer where I can contribute to innovative engineering

TECHNICAL SKILLS

Programming Languages

- Python
- JavaScript
- Java
- C#
- C++
- PHP
- SQL
- HTML5
- CSS3

Frameworks & Libraries

- React
- Flask
- Node.js

AI & Automation

- OpenAI API
- LangGraph
- AI Agents
- Prompt Engineering
- Semantic Search (FAISS)
- Process Automation
- Email Automation

Backend Development

- REST API Development
- Authentication
- SQLite
- MySQL
- Database Design

Tools & Technologies

- Git
- GitHub
- Docker
- Linux
- Visual Studio Code

Software Engineering

- Object-Oriented Programming
- Agile Development
- Software Testing
- Debugging
- System Design
- Problem Solving

FEATURED TECHNICAL PROJECTS

AI Recruitment Agent

Technologies: Python • Flask • LangGraph • OpenAI API • SQLite • FAISS • Docker

Designed and developed an intelligent AI-powered recruitment platform that automates the candidate screening and hiring workflow using multiple AI agents.

Key Features

- Automated CV parsing and information extraction
- AI-powered candidate ranking against job requirements
- Semantic candidate search using FAISS vector embeddings
- Intelligent job matching and recommendation engine
- Automated interview invitation and rejection emails
- PDF report generation for recruiters
- Modular multi-agent workflow using LangGraph
- RESTful backend architecture

Skills Demonstrated

- Backend Development
- Artificial Intelligence
- Prompt Engineering
- API Integration
- Database Management
- Software Architecture
- Process Automation

Personal Developer Portfolio

Technologies: React • JavaScript • HTML • CSS • Vite

Designed and developed a fully responsive portfolio website showcasing software development projects, technical skills, and professional experience using modern UI principles, reusable React components, and animated user interactions.

PROFESSIONAL EXPERIENCE

Agentic AI Software Developer Intern

New Island Technologies

April 2026 – Present

Responsibilities & Achievements

- Designed and developed intelligent agent-based software solutions using modern AI development frameworks and Python.
- Built and integrated AI agents capable of reasoning, task execution, workflow automation, and decision-making.
- Developed scalable backend services and RESTful APIs to support AI-powered applications.
- Implemented prompt engineering techniques to improve AI response quality, accuracy, and reliability.
- Worked with Large Language Models (LLMs) to automate business processes and enhance user experiences.
- Designed modular software architectures following software engineering best practices for maintainability and scalability.
- Integrated databases and external APIs into AI workflows for intelligent data processing and automation.
- Collaborated within an Agile development environment using Git for version control and team collaboration.
- Participated in debugging, testing, and optimizing AI applications to improve performance and reliability.
- Contributed to the development of production-ready AI solutions by following clean code principles and modern software engineering practices.

Technical Skills Used

Python • React • Node.js • Flask • LangGraph • OpenAI API • REST APIs • SQL • Git • Docker • Prompt Engineering • AI Agents • Automation • Agile

Software Developer Intern

Dev Shack

July 2025 – November 2025

Responsibilities

- Developed production-ready full-stack functionality using React.js and Node.js.
- Integrated REST APIs and authentication systems to improve application security and functionality.
- Collaborated within Agile development teams to deliver software features under project deadlines.
- Utilised Git for version control and Docker to streamline development and deployment workflows.
- Participated in debugging, testing, and continuous improvement of existing applications.
- Contributed to designing scalable and maintainable software solutions.

IT Support Technician

Richfield Graduate Institute

February 2024 – November 2024

Responsibilities

- Provided first-line technical support for staff and students.
- Diagnosed and resolved hardware, software, and networking issues.
- Maintained computer laboratories and institutional IT infrastructure.
- Improved system availability through proactive troubleshooting and maintenance.
- Assisted users with software installation, configuration, and technical guidance.

Software Engineering Tutor

Richfield Graduate Institute

2023 – 2024

Responsibilities

- Tutored students in Java, Python, C++, C#, and Software Engineering principles.
- Assisted students with debugging, assignments, and practical programming projects.
- Improved student understanding through one-on-one mentoring and structured guidance.
- Supported software design and object-oriented programming concepts.

EDUCATION

Bachelor of Science in Information Technology

Richfield Graduate Institute

2022 – 2024

Academic Average: 73%

LEADERSHIP & ACHIEVEMENTS

- Developed an AI-powered Recruitment Agent demonstrating AI, backend development, and automation skills.
- Hackathon Participant (Team Leader).
- Deputy Head Girl (Representative Council of Learners).
- Successfully balanced academic studies with tutoring and technical support responsibilities.

CORE COMPETENCIES

- Software Development
- Backend Development
- Full-Stack Development
- Artificial Intelligence
- REST API Development
- Database Design
- Process Automation
- AI Agent Development
- Git Version Control
- Docker
- Agile Development
- Problem Solving
- Team Collaboration
- Technical Troubleshooting
- Software Testing